

VLAN Port Types: Trunk vs Access

Access Port

- Each Port assigned to a single VLAN
- Carries untagged packets sent to or received from connected devices
- Switch adds VLAN tag based on port configuration
- No VLAN knowledge required for end devices
- Single broadcast domain per VLAN

Trunk Port

- Used to connect to other switches for inter-switch communication
- Carries tagged packets that belong to multiple VLANS
- Enforces VLAN isolation
- Supports up to 4094 VLANS(802.1Q standard)

Untagged Packet

MAC Header	Data	CRC
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Standard Ethernet II Frame

- No VLAN information
- Typically 64-1518
- Used by end devices (computers, phones, etc)

Tagged Packet (802.1Q)

MAC Header	VLAN Tag	Data	CRC
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TPID 0x8100	PCP Priority (0-7)	VLAN ID 1-4094	Ethernet Original Type
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- 4-byte VLAN tag added (total frame 60-1522 bytes)
- QoS priority can be set with Priority Code Point (PCP) field

Example Network Diagram

